

Chapter 54 — Management of Paediatric Kidney Stone Disease

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EPIDEMIOLOGY OF EARLY-ONSET KIDNEY STONE DISEASE

- **Nephrolithiasis is a disorder of mineral metabolism** punctuated by acute symptomatic events; **about 50% recur within 5 years** of the index presentation.
- **United States (US) prevalence has doubled:** 5% (1988–1994) → **11% (2015–2018)**.
- **Incidence in youth is rising 5–10% per year**, disproportionately to adults; **the greatest rise is in adolescents, particularly females** — in females aged 15–19 years, **75 → 120 per 100,000** (1997 → 2012), **52% higher than in males** of the same age.
- **Sex ratio reverses with age of onset:** adult-onset disease was historically **male-predominant, 3 : 1** (gap now narrowing); early-onset disease is **female-predominant**. The reason is unknown.
- **Ethnicity:** commonest in non-Hispanic white children → Hispanic white → African American.
- **Early-onset disease** is defined as onset at **<24 years**.
- **Burden:** rising hospitalisations, emergency department (ED) visits, operations and costs — **at least US\$375 million per year** (2010) in hospital and ED care alone, before outpatient care, drugs, imaging and caregivers' lost earnings. **Recurrence turns a single event into a chronic condition**.

STONE COMPOSITION — ADULTS VERSUS CHILDREN

Predominant composition	Adults	Children
Calcium oxalate (CaOx)	75–80%	Similar (about 80%)
Calcium phosphate (CaP)	About 5%	Slightly more common
Struvite	10–20%	Not separately stated
Pure uric acid	About 5%	Less common

- **Children's urine is more alkaline**, which favours CaP and disfavours uric acid (which precipitates in acid urine).
- **The view that childhood stones are mostly genetic, inborn-error or infective is no longer true;** most are idiopathic CaOx, as in adults.
- **Randall's plaques — hydroxyapatite (calcium phosphate) deposits on the renal papillae — are the platform on which most idiopathic CaOx stones grow;** children with CaOx stones show **the same papillary changes as adults** (white papillary deposits to which stones attach).

Determinants

- **Genetic susceptibility × exposome** (non-genetic environmental exposures) × **comorbid disease** — rare monogenic disease excepted.
- **Heritability is about half, higher in males** (twin studies); it is **not explained by rare monogenic disease** (primary hyperoxaluria, cystinuria). Polygenic determinants remain undefined, though polymorphisms raise risk both in healthy people and in those with diabetes, obesity and gout.
- **Incomplete genetic determination implies gene-exposome interaction** — diet, medications, ambient environment. **Low fluid and high sodium intake** are the two key Western-diet risks.
- **Medications act by two routes:**
 - **Direct** — the drug crystallises through poor urinary solubility: **1–2% of stones**.

- **Indirect — antibiotics:** oral **cephalosporins, broad-spectrum penicillins, nitrofurantoin, fluoroquinolones and sulfonamides** raise the odds of stones (population-based case-control study); **risk wanes but persists up to 5 years and is greatest in children**; the same association is seen in adult women.
- **Proposed mechanism — microbiome disruption:** children with CaOx stones have **fewer oxalate-degrading and butyrate-producing taxa**. Butyrate is anti-inflammatory, protects against renal ischaemic injury, maintains the gut mucosal barrier and **regulates intestinal SLC26 oxalate transporters**; losing both functions **synergistically increases oxalate absorption and excretion**.

HIGH-RISK GROUPS

Group	Mechanism of stone formation
Bowel-segment reconstruction (augmentation cystoplasty): spina bifida about 1% per year, $\geq 10 \times$ general population	Metabolic acidosis; bowel mucus as nidus; bacterial colonisation; hyperoxaluria from malabsorption after bowel resection
Malabsorption; urinary stasis (anomalies, immobility)	Enteric hyperoxaluria; stasis
Enteral tube feeds	Feeds high in sodium and oxalate, low in calcium → ↑ calciuria and ↑ gut oxalate absorption; hyperoxaluria commoner than in ambulatory stone formers
Non-ambulatory children	↑ bone resorption, ↓ osteoblast activity → hypercalciuria; more often need surgery; more infection stones (calcium carbonate, struvite)

A Disorder of Mineral Metabolism

- **About 50% recur within 3–5 years** of the index stone — **equal to or higher than in adults**; **adolescents have more symptomatic events than younger children**.

EXTRARENAL ASSOCIATIONS (MOSTLY ADULT DATA)

Outcome	Association	Proposed mechanism
Chronic kidney disease (CKD)	↑ risk; end-stage renal disease (ESRD) roughly doubled after ≥ 1 episode; rises with number of episodes	—
Hypertension	↑ risk, similar in size to adult estimates (Health Professionals Follow-Up Study; Nurses' Health Study I and II)	Altered renal calcium and sodium handling
Coronary heart disease	↑ in women, relatively greater in younger women	—
Fracture / low bone mineral density (BMD)	↑ fracture risk, greatest at 10–19 years	Hypercalciuria, negative calcium balance

- **Childhood is the critical window for bone-mass accrual** → identify low BMD early; **citrate and thiazides may reduce adult osteoporosis risk**.
- **Children may be at particular risk:** early onset may mark a **more severe phenotype**, and they have a **longer lifetime** in which comorbidity can develop.

KEY POINTS: EPIDEMIOLOGY OF KIDNEY STONE DISEASE

- **Dramatic rise in paediatric incidence**, particularly in adolescent females; **female predominance** in early-onset disease.
- **About 80% of paediatric stones are predominantly CaOx**; **CaP slightly commoner**, **pure uric acid less common** than in adults.
- **Genetic susceptibility, the exposome and comorbid disease** are the primary determinants.

- A **disorder of mineral metabolism**, associated with **cardiovascular disease, CKD and decreased BMD**.

Concept — *Early-onset stone disease is the adult disease started early: the same idiopathic CaOx-on-Randall's-plaque pathway, driven by gene-exposome interaction, with a longer runway for recurrence, CKD, hypertension and bone loss.*

EVALUATION

Child with Suspected Nephrolithiasis

- **Older children present as adults do:** flank pain, nausea and vomiting, lower urinary tract symptoms.
- History: onset, frequency and duration of pain; personal and family stone history; relevant urological surgery; factors raising pre-test probability. **Always ascertain fever.**
- **Young or non-verbal children cannot localise pain** → rely heavily on imaging and laboratory tests.

Imaging

- **European Association of Urology (EAU) and an American Urological Association (AUA) technical paper agree: ultrasound (US) first; non-contrast computed tomography (NCCT) only if US is non-diagnostic and suspicion remains high.**
- **Image Gently Alliance (2007)** — campaign to lower paediatric radiation; **US-first pathways reduce initial CT use.**
- **Diagnostic algorithm:** suspected stone → **US (grey-scale + colour Doppler)** → positive → treat as indicated; negative + **high suspicion ± secondary signs (e.g. hydronephrosis)** → **low-dose NCCT**; negative + low suspicion → stop.
- **US stone criteria (both): hyperechoic focus in papilla, calyx or pelvis → confirmatory twinkle artefact on colour Doppler.**
- **Maximise Doppler pulse repetition frequency (PRF)** to suppress vascular-flow noise.
- **Posterior acoustic shadow** greatly raises specificity when present, but **modern harmonic and compound imaging rarely shadows stones <4 mm.**
- **Stones missed on US are usually clinically insignificant** (small or non-obstructing). Renal resistive indices have been studied in adults as an adjunct.
- **US resolution exceeds many CT scans:** twinkling foci of **about 2 mm** invisible on CT are often **Randall's plaques ± small attached stones** (CT shows only hyperdense papillae) — consistent with **US overestimating stone size by 1–2 mm.** Plaques share the US characteristics of stones but are typically **<4 mm.**

ULTRASOUND VERSUS NON-CONTRAST CT

	Ultrasound	Non-contrast CT
Sensitivity / specificity	>70% / >95%	About 100% / about 100% (gold standard)
Radiation	None	Ionising — small risk per scan, cumulative
Ureteric stones	Poorly visualised	Confirms or excludes
Foci about 2 mm	Detected (often plaques)	Not seen as distinct calculi
Size	Overestimates by 1–2 mm	Reference
Role	First-line	Second-line: US non-diagnostic + high suspicion

- **Children carry greater CT risk — longer life expectancy, radiosensitive tissue;** a substantial minority have ≥2 CTs per episode.

- **When CT:** high suspicion + non-diagnostic US (e.g. hydronephrosis without visible stone) → **low-dose NCCT of abdomen and pelvis** → size and location of a ureteric stone decide observation versus surgery → **prevents silent obstruction and loss of renal function.**
- **Dose reduction:** 40 mA (half standard) for children ≤50 kg keeps diagnostic accuracy; much lower amperage still detects 2–8 mm stones at about one-third of the dose.

Concept — Paediatric imaging is governed by **radiation economy**: US is accepted despite imperfect sensitivity because what it misses is usually insignificant, and CT is reserved for the one situation — a suspected obstructing ureteric stone not seen on US — where a miss would cost renal function.

Child with Confirmed Nephrolithiasis

Medical History and Physical Examination

- **Family history of stones, haematuria or CKD** → inherited disease: **cystinuria, primary hyperoxaluria, CYP24A1 mutations, Dent disease.**
- **Dietary history in every child:** fluid, salt, vitamins C and D, mineral supplements, special diets (**ketogenic**).
- **Medication history:** corticosteroids; **furosemide, acetazolamide; indinavir;** antibiotics; **topiramate, zonisamide.**
- **At-risk history:** prematurity, urinary tract anomalies, urinary tract infection (UTI), malabsorption, prolonged immobility.
- **Examination clues:** dysmorphism → **Williams syndrome**; rickets → **Dent disease**, hereditary hypophosphataemic rickets with hypercalciuria (HHRH); tetany → familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (FHHNC), autosomal dominant hypocalcaemic hypercalciuria (ADHH); gout → hypoxanthine–guanine phosphoribosyltransferase (HPRT) deficiency, phosphoribosyl pyrophosphate synthetase (PRPS) superactivity.

Metabolic Evaluation

- **An identifiable metabolic abnormality raises recurrence risk up to fivefold** → every child has a comprehensive metabolic evaluation.
- **Rationale in early onset:** more autosomal recessive (monogenic) causes; a possibly more severe phenotype; a longer time for recurrence and end-organ effects (e.g. BMD).
- **Start with stone analysis** — **infrared spectroscopy or X-ray diffraction**; **cystine, struvite, 2,8-dihydroxyadenine (2,8-DHA) and xanthine** are diagnostic in themselves.

SERUM AND SCREENING STUDIES

Test	Purpose / when
Creatinine	Essential — acute kidney injury or CKD
Calcium, phosphorus, bicarbonate, magnesium, uric acid	Hyper/hypocalcaemic stones; hyper/hypouricaemia; normocalcaemic hypercalciuria disorders — HHRH, Bartter syndrome, distal renal tubular acidosis (dRTA), FHHNC
Intact parathyroid hormone (PTH)	Not routine — primary hyperparathyroidism is rare in children ; only with hypercalcaemia + hypophosphataemia
25-hydroxyvitamin D	Hypercalcaemia and/or hyperphosphataemia with a relevant exposure
25-hydroxyvitamin D : 24,25-dihydroxyvitamin D (normal <25–30)	CYP24A1 screen — hypercalcaemia + hypercalciuria after other causes excluded. Biallelic loss of function → no peripheral vitamin D catabolism → persistently high 1,25-dihydroxyvitamin D, mimicking vitamin D intoxication

Test	Purpose / when
Spot urine β2-microglobulin or retinol-binding protein	Dent disease screen — males with recurrent calcium stones plus proteinuria, or family history of renal failure, focal segmental glomerulosclerosis or recurrent stones

- **24-hour urine:** collect → analyse **calcium, oxalate, uric acid, sodium, citrate, creatinine, volume, pH, cystine** (cyanide–nitroprusside screen) → **index to weight, body surface area (BSA) or creatinine** → check adequacy by **creatinine excretion** (adult 15–20 mg/kg/day) → compute supersaturations (CaOx, CaP, uric acid).
- **One or two collections — debated:** one may suffice; two give more specific diagnoses.
- **Not toilet-trained → spot solute : creatinine ratios** (less precise). Repeat weeks to months after a diet change or new drug.
- **Crystalluria is generally non-diagnostic, except:** hexagonal → cystine; coffin-lid (triple phosphate) → struvite; brown spiculated → 2,8-DHA.

URINARY METABOLIC ABNORMALITIES

Hypercalciuria

- **Commonest abnormality:** 30–50% of stone-forming children. **Commonest cause at all ages: idiopathic hypercalciuria** — hypercalciuria **without hypercalcaemia** and no other cause.
- **Familial idiopathic hypercalciuria:** gene(s) unknown; **autosomal dominant, incomplete penetrance**. Present in **about 4% of healthy children, 40–50% of whom have a family history** of stones.
- **Definition:** **>4 mg/kg/day in children >2 years**, or total **>200–250 mg/day**. Not toilet-trained → random calcium : creatinine; **school-age normal ≤0.21 mg/mg**; physiological excretion is higher in younger children.
- **Sequence:** confirm hypercalciuria → **check serum calcium** → exclude secondary causes → only then diagnose idiopathic hypercalciuria.

HYPERCALCIURIA CLASSIFIED BY SERUM CALCIUM

Serum calcium	Causes to exclude
High	Hyperparathyroidism; biallelic CYP24A1; hypervitaminosis D; when indicated — immobilisation, sarcoidosis, malignancy, juvenile idiopathic arthritis, corticosteroid excess, adrenal insufficiency, Williams syndrome
Low	Hypoparathyroidism; ADHH (gain-of-function calcium-sensing receptor)
Normal (most → idiopathic)	Prematurity; furosemide, acetazolamide; topiramate; ketogenic diet; Dent; Bartter; FHHNC; dRTA; HHRH; possibly medullary sponge kidney

Concept — Hypercalciuria is a sign, not a diagnosis: serum calcium sorts it into three bins, and idiopathic hypercalciuria is named only once the normocalcaemic bin has been cleared of secondary causes.

TABLE 54.1 NORMAL VALUES FOR URINARY SOLUTE EXCRETION

Metabolite	Random, indexed to creatinine (mg/mg), by age	24-h indexed (all ages)	24-h total
Calcium	0–6 mo <0.8; 7–12 mo <0.6; >24 mo <0.21	<4 mg/kg	Male <250; female <200 mg/day
Oxalate	0–6 mo <0.26; 7–24 mo <0.11; 2–5 y <0.08; 5–14 y <0.06; >16 y <0.03	<50 mg/1.73 m ² BSA	20–40 mg/day
Citrate (lower limit)	0–5 y and >5 y: >0.2–0.42	Male >180; female >300 mg/g creatinine	Male >450; female >550 mg/day

Metabolite	Random, indexed to creatinine (mg/mg), by age	24-h indexed (all ages)	24-h total
Cystine	>6 mo <0.075	<50 mg/1.73 m ² BSA	—
Uric acid	>2 y <0.56 mg/dL GFR	<815 mg/1.73 m ² BSA	—

BSA, body surface area; GFR, glomerular filtration rate.

- **Purpose:** paediatric urine chemistry must be **indexed to age, weight, BSA or creatinine**, never read against adult absolute limits.
- **Pattern:** random calcium and oxalate ratios fall with age; citrate is the only lower limit — an inhibitor, so the abnormality is too little.
- **Traps:** calling an infant hypercalciuric on the school-age 0.21 ratio; forgetting citrate thresholds are minimums.
- **The body text gives different reference values for three endpoints:** male hypocitraturia <128 mg/g (table: normal >180); normal oxalate <40 mg/day/1.73 m² (table <50); normal cystine <60 mg/day/1.73 m² (table <50).

Hyperoxaluria

- Oxalate is an end product of glyoxylate and ascorbic acid metabolism, renally excreted; **80–85% endogenous, 10–15% dietary**. Normal <40 mg/day/1.73 m²; young children → random oxalate : creatinine.
- **Primary = inherited hepatic overproduction** (rare) versus **secondary = increased absorption or excess precursor intake** (commoner).

Primary Hyperoxaluria (PH)

- **PH1–3:** rare autosomal recessive disorders of hepatic oxalate overproduction → hyperoxaluria → nephrocalcinosis and nephrolithiasis → progressive CKD.
- **Phenotype spans neonatal ESRD to occasional stone passage in adulthood → diagnosis often delayed, sometimes until loss of a transplanted kidney.**

PRIMARY HYPEROXALURIA TYPES COMPARED

	PH1	PH2	PH3
Gene	AGXT	GRHPR	HOGA1
Enzyme	Alanine-glyoxylate aminotransferase (AGT), hepatic peroxisomal; pyridoxine cofactor	Glyoxylate reductase-hydroxypyruvate reductase	Mitochondrial 4-hydroxy-2-oxoglutarate aldolase (hydroxyproline pathway, 4th step)
Urinary marker	↑ Glycolate (and glyoxylate)	↑ L-glycerate	↑ 4-hydroxy-2-oxoglutarate
Severity	Most severe; profound B6 deficiency mimics it	Milder, but not benign — can reach ESRD	—

- **Sequence:** raised 24-h oxalate → urinary glycolate, glycerate, 4-hydroxy-2-oxoglutarate → **genetic testing**. Raised oxalate with all three normal → **secondary hyperoxaluria**.

Secondary Hyperoxaluria

- **Excess intake** of oxalate or precursors (**high-dose vitamin C**) versus **enteric hyperabsorption**.
- **Gut oxalate absorption varies inversely with dietary calcium → calcium-deficient diets raise urinary oxalate.**
- **Enteric causes** (fat malabsorption, chronic diarrhoea): inflammatory bowel disease, coeliac disease, exocrine pancreatic insufficiency (cystic fibrosis), biliary disease, small-bowel resection / short bowel.

- **Mechanism:** free fatty acids **bind luminal calcium** → more **unbound oxalate** absorbed.

Hypocitraturia

- Citrate is reabsorbed and metabolised in the **proximal tubule**. **Defined (24 h) as citrate : creatinine <128 mg/g (males), <300 mg/g (females).**
- **Mechanism:** **proximal tubular intracellular acidosis** (metabolic acidosis or hypokalaemia) → ↑ citrate reabsorption.
- **Causes:** ketogenic diet; **topiramate, acetazolamide; dRTA**; chronic diarrhoea. **Incomplete dRTA has no overt acidosis or hypokalaemia** → missed unless **acid-load testing** is done.
- **Most cases are idiopathic;** diets **high in animal protein, low in fibre and potassium** lower citrate.

Cystinuria

- **Autosomal recessive; SLC3A1 or SLC7A9** → failed proximal reabsorption of **cystine, lysine, ornithine, arginine** — only cystine crystallises.
- **Normal <60 mg/day/1.73 m²; homozygotes often >400.**
- Usually presents in the **second or third decade**, but **can present in infancy with staghorn calculi**, occasionally bladder stones.
- **Poor solubility** → recurrent stones → **long-term renal failure risk**; UTI; **mixed cystine/struvite** stones.

Hyperuricosuria

- **Children excrete more uric acid:** fractional excretion **30–50% (neonate) → 8–12% (adult level)** by adolescence. **Paradox: uric acid stones are <5% of paediatric calculi.**
- **Epitaxy:** one crystal lattice (uric acid) seeds growth of another (CaOx) in the same orientation → **hyperuricosuric CaOx stones.**
- **Definition >815 mg/day/1.73 m²;** per-GFR excretion is constant after age 2. **Spot urine (>2 years): >0.56 mg/dL** $GFR = \text{urine uric acid} \times \text{plasma creatinine} \div \text{urine creatinine}$ (all mg/dL).
- **Greatest risk for uric acid stones: hyperuricosuria + low urine pH.**

HYPERURICOSURIA BY SERUM URATE

Serum urate	Causes
High	HPRT deficiency (Lesch–Nyhan syndrome), PRPS superactivity; lymphoproliferative disorders; polycythaemia
Low	Hereditary renal hypouricaemia — SLC22A12 or SLC2A9 (proximal urate transporters): stones and exercise-induced acute renal failure
Not specified	Purine excess (animal protein, anchovies, mussels); haemolysis; probenecid, salicylates, losartan; cyanotic congenital heart disease; melamine; idiopathic (familial)

TABLE 54.2 MONOGENIC CAUSES OF KIDNEY STONE DISEASE

Disease	Inheritance	Gene(s)	Signature features
Normocalcaemic hypercalciuria			
Dent disease	X-linked	CLCN5	Low-molecular-weight proteinuria , nephrocalcinosis, hypercalciuria, CKD
Bartter syndrome	AR	Many	Salt wasting, polyuria, hypokalaemia, metabolic alkalosis , hypercalciuria, ± sensorineural deafness
FHHNC	AR	CLDN16, CLDN19	Hypomagnesaemia, seizures/tetany , hypercalciuria, nephrocalcinosis

Disease	Inheritance	Gene(s)	Signature features
Distal RTA	AD, AR	SLC4A1; ATP6V1B1; ATP6V0A4	Hypocitraturia, metabolic acidosis , hypokalaemia, nephrocalcinosis
Other monogenic causes			
Primary hyperoxaluria	AR	AGXT, GRHPR, HOGA1	Hyperoxaluria, nephrocalcinosis, CKD, oxalosis
Cystinuria	AR	SLC3A1, SLC7A9	Dibasic aminoaciduria; heterogeneous
HPRT deficiency	X-linked	HPRT1	Developmental delay ; hyperuricaemia, hyperuricosuria
PRPS superactivity	X-linked	PRPS1	Hyperuricaemia, gout, hyperuricosuria, uric acid stones

AD, autosomal dominant; AR, autosomal recessive.

- **Purpose:** recognising the monogenic child — the rationale for evaluating every early-onset stone former.
- **Pattern:** autosomal recessive dominates; the X-linked trio (Dent, HPRT, PRPS) presents in boys; every normocalcaemic-hypercalciuria disorder causes nephrocalcinosis.
- **Traps:** Bartter = alkalosis, dRTA = acidosis (both hypokalaemic and hypercalciuric); Dent disease is detected by low-molecular-weight proteinuria, not serum calcium.

KEY POINTS: EVALUATION OF CHILDREN WITH KIDNEY STONE DISEASE

- An identifiable metabolic abnormality increases recurrence risk.
- Every child needs a comprehensive evaluation, including 24-hour urine for calcium, oxalate, uric acid, sodium, citrate, creatinine, volume, pH and cystine.
- Idiopathic hypercalciuria in about 30–50% of stone-forming children.
- Hypocitraturia, hyperoxaluria and cystinuria remain important and must be identified.

Concept — Every urinary risk factor is either an excess of a promoter (calcium, oxalate, urate, cystine) or a deficit of an inhibitor (citrate), each with a rare monogenic cause at one end and a common idiopathic or dietary cause at the other — hence the work-up runs stone analysis → serum → indexed urine → specific markers or genetics.

MANAGEMENT OF CHILDREN AND ADOLESCENTS WITH KIDNEY AND URETERAL STONES

Medical Expulsive Therapy

- **Options for a ureteric stone:** observation with analgesia → observation plus **medical expulsive therapy (MET)** → surgery.
- **MET = α -blockers** (less often calcium-channel blockers): **α 1A/ α 1D antagonism** in the **distal-third ureter and vesicoureteric junction** → ureteric dilatation.
- **Spontaneous passage is higher in adults, for stones <5 mm and for distal stones.** α -Blockers help most in adults with distal stones >5 mm; the paediatric size of greatest benefit is unknown.
- **AUA / Endourological Society (EUS) 2016:** uncomplicated ureteric stones ≤ 10 mm → **observation $\pm$ α -blocker MET** (moderate; grade B). A later meta-analysis of randomised trials confirmed **α -blockers increase passage in children.**
- **Counsel:** off-label; maximum duration unknown but **≤ 6 weeks**; tamsulosin appears effective and safe.

Surgical Management

- **Up to 60% of children with renal or ureteric stones need surgery, yet nephrolithiasis has the fewest randomised trials of ten major kidney diseases.**
- **Options as in adults — ureteroscopy (URS), shockwave lithotripsy (SWL), percutaneous nephrolithotomy (PCNL) — all need anaesthesia and, traditionally, radiation.**
- **Ideal modality:** (1) high monotherapeutic success; (2) low complication risk; (3) return to baseline activity; (4) minimal radiation; (5) minimal anaesthesia; (6) no ancillary procedures.
- **Patient-centred choice:** patient and stone factors, surgeon experience, resources — and, **most importantly, patient-driven goals of care.**
- **Only 8 AUA/EUS 2016 recommendations address children; half are expert opinion.** The **2017 National Urology Research Agenda** prioritised paediatric stones; the **Pediatric KIDney Stone (PKIDS) Care Improvement Network** is comparing clinical and patient-reported outcomes of URS, SWL and PCNL.

TABLE 54.3 AMERICAN UROLOGICAL ASSOCIATION AND ENDOUROLOGICAL SOCIETY 2016 GUIDELINES FOR THE SURGICAL MANAGEMENT OF STONES IN PEDIATRIC PATIENTS

Scenario	Recommendation	Strength	Evidence
Uncomplicated ureteric stone ≤10 mm	Observation ± α-blocker MET	Moderate	B
Ureteric stone unlikely to pass / failed observation or MET	URS or SWL by anatomy and habitus	Strong	B
Ureteric stone before URS	No routine pre-stenting	Expert opinion	—
Renal burden ≤20 mm	SWL or URS first-line	Moderate	C
Renal burden >20 mm	PCNL or SWL; if SWL → stent or nephrostomy	Expert opinion	—
Before PCNL	Low-dose CT	Strong	C
Upper tract stones, no anatomic abnormality	No routine open/laparoscopic/robotic surgery	Expert opinion	—
Asymptomatic, non-obstructing renal stones	Active surveillance with periodic US	Expert opinion	—

Data from Assimos et al., 2016.

- **Purpose:** the whole paediatric surgical decision tree in eight lines.
- **Pattern:** two size cut-offs — 10 mm (ureteric) and 20 mm (renal); SWL appears in every active renal row; only two recommendations are strong.
- **Traps:** strength ≠ evidence (CT before PCNL is strong on grade C evidence); confusing the 10 mm ureteric and 20 mm renal thresholds.

Surgical Antimicrobial Prophylaxis

- **All URS and PCNL; SWL only if infection risk is raised.** No paediatric-specific recommendations.
- **Urine culture before every upper tract procedure** — guides antibiotics, especially for percutaneous surgery, high-grade obstruction and indwelling stents.
- **Authors' regimen:** negative culture, uncomplicated URS → **perioperative cefazolin**; percutaneous surgery or existing stent/nephrostomy → **fluoroquinolone or ampicillin + gentamicin. Stop within 24 hours** unless infection is suspected.

GOALS OF THERAPY

- **As in adults: stone clearance with the least harm** to physical, emotional and social health.

- **Primary determinants:** stone size and location, anatomy, patient (and provider) preference; **secondary:** comorbidity, composition, equipment. **In practice the treating site strongly shapes the choice.**

Stone Clearance

- **The most important surgical outcome;** residual fragments → growth and symptomatic events.
- **No consensus on significant residual fragment size in children;** assessment (visual at URS, US, CT) varies, hampering comparison.
- **Why 4 mm on US: US overestimates by about 2 mm + 2 mm = conventional CT threshold** for insignificance.

Radiation

- **Children face more future imaging and interventions,** and higher cumulative risk (longer life, radiosensitive tissue). Exposure spans diagnosis, operation and surveillance.
- **URS and PCNL use fluoroscopy; SWL does unless US-targeted. PCNL delivers several times the dose of URS;** bilateral URS more than unilateral.
- **Limits: ≤50 mSv in any 12 months; average <20 mSv/year over 5 years.**
- **Reduction levers:** maximise source-to-skin distance, improve skin-to-intensifier distance, collimate; lower kVp, pulsed dosing, paediatric dose settings, **no routine magnification; intermittent low-dose rather than continuous fluoroscopy; radiation checklists; US guidance for URS, PCNL access and SWL targeting.**

Ancillary Procedures / Anaesthesia

- **Several procedures and anaesthetics are often needed:** a small ureter may force a **pre-stent for passive dilatation** (an extra anaesthetic); SWL often needs retreatment.
- **Concern: anaesthetic neurotoxicity in the developing brain, especially pre-school children.**

Concept — *Paediatric stone surgery is judged on a ledger with more columns than in adults: clearance must be bought with the fewest anaesthetics and the least cumulative radiation, because the child carries both costs across a longer lifetime.*

URETEROSCOPIC MANAGEMENT OF UPPER URINARY TRACT CALCULI

- **URS is a mainstay:** pooled paediatric series show **clearance approaching 90%** with mostly low-grade complications. **Thulium fibre laser cleared better than low-power holmium,** without more complications.
- **Efficacy falls with lower-pole location and small patient size:** small kidneys and **tight infundibulopelvic angles** limit manoeuvring; **children <6 years** have lower success and more complications; **<20 kg** carries a high complication rate.
- **Pre-stenting.** Forcing a scope or **active dilatation risks perforation and stricture**, so some ureters need a stent first; but **AUA/EUS advise against routine pre-stenting**, which is linked to **more ED visits after URS.** **α-Blockers may aid primary access** (preliminary).

LASER DUSTING VERSUS FRAGMENTATION-EXTRACTION

	Dusting	Fragmentation and extraction
Settings	High frequency (>50 Hz), low energy (<0.4 J)	Low frequency, high energy
Fragments	<1 mm — pass spontaneously	1–4 mm — extracted
Adult outcomes	Lower clearance, less stenting, similar unplanned encounters	—

	Dusting	Fragmentation and extraction
Drawback	—	Multiple ureteric passes → access sheath

- **Ureteral access sheath (UAS).** **Benefits:** repeated access, ↓ intrarenal pressure, ↓ operative time, easier flexible URS in tortuous or altered anatomy; inner 9–12 Fr, outer 11–14 Fr. **Risk:** some ureters cannot take a sheath → injury. Paediatric clearance benefit unproven.

Equipment

SEMIRIGID VERSUS FLEXIBLE URETEROSCOPES

	Semirigid	Flexible
Outer size	4.5–6 Fr	7.5–9 Fr
Working channel	2.4–3.5 Fr	1.8–3.6 Fr
Deflection	—	Up to 270° → most lower-pole stones; reduced by laser fibre and larger fibres, worst in small children
Use	Ureteric stones	Proximal ureteric and renal stones

- Optics fibreoptic or digital. Wires 0.035, 0.025, 0.018 inch; stents 3.7–8 Fr.
- **The working channel dictates compatible wires and catheters: a 4 Fr access catheter or 3.7 Fr stent takes only a 0.025-inch wire.**
- **Basic paediatric endourology suite:** ureteroscopes — 7.5–8 Fr flexible, 4.5 and 6.5 Fr semirigid; guidewires — 0.035-inch floppy hydrophilic tip and 0.018–0.025-inch hydrophilic (each straight and angled), 0.038-inch stiff; open-ended ureteric catheters 3, 4, 5 Fr and dual-lumen catheter; holmium and/or thulium laser; 4-wire and/or 16-wire baskets; 8/10 Fr coaxial dilators; access sheaths 9.5 and 12 Fr internal diameter; stents 3.7, 4.6–4.7, 6 and 8 Fr.

Ureteroscopic Technique in Children

Sequence: general anaesthesia → **confirm sterile urine culture** → lithotomy (sized stirrups; **infants: gel rolls under the knees**; pad pressure points; **leg weight on heels, not calves or knees — peroneal nerve**) → broad-spectrum prophylaxis → rigid cystoscopy (7.5, 10 or 14.5 Fr) → **retrograde pyelogram (RPG) + wire under fluoroscopy** → calibrate orifice; **safety wire via 8/10 Fr dual-lumen catheter/coaxial dilator** → **if 8/10 Fr will not pass: 4.6–4.8 Fr stent for ≥14 days**, then return → URS.

- **No balloon dilatation of the orifice (stricture); no active distal dilatation** — injury risk, and the ureter proximal to the orifice also limits safe URS.
- **Semirigid for ureteric, flexible for proximal ureteric and renal stones — both over a safety wire. UAS (10/12 or 12/14 Fr) routine** for flexible URS, especially large proximal ureteric stones and renal burdens needing basketing.
- **URS is less position-dependent than SWL**, but the small paediatric lower pole is still hard to reach: **150–200 µm fibres for the lower pole** (better deflection) versus **365 µm for ureteric, mid- and upper-pole stones**.
- **Irrigant isotonic, at body temperature** → no hyponatraemia or hypothermia. **Basket when feasible**; otherwise **holmium:YAG** fragmentation.
- **Postoperative stent decision: duration, number of scope passes, visible ureteric trauma or oedema.** Removal: **string 5–7 days** (home or nurse) versus **cystoscopy under anaesthesia after 7 days**.
- **Stent symptoms are common** → paracetamol, ibuprofen, α-blockers, anticholinergics; **no routine opioids**.

Limitations and Complications

- **Ureteric injury, UTI, bleeding; >Clavien III uncommon.**
- **Ureteric injury → abort and stent** (limits shear, ischaemia, extravasation). Injury arises at scope introduction, during instrument passage — **especially at an impacted stone** — and when **withdrawing a basketed stone too large for the ureter.**
- **Stricture:** RPG defines degree, length and location → short mid-ureteric: prolonged stenting (uncertain long-term) · endoscopic: **laser incision, not balloon** · longer, denser or recurrent: **formal (often robotic) reconstruction** · distal: **ureteroneocystostomy** · proximal: **pyeloplasty** · mid: **excision and primary anastomosis.**

KEY POINTS: URETEROSCOPIC MANAGEMENT OF UPPER TRACT CALCULI

- Miniaturised instruments and adapted adult techniques treat calculi <15 mm effectively and safely.
- No routine pre-stenting.
- Postoperative stenting depends on duration, number of passes and visible trauma or oedema.

Concept — Every paediatric ureteroscopy decision is a negotiation with calibre: **the ureter's size decides pre-stenting, wire, sheath and scope, and the kidney's size decides fibre choice** — so passive dilatation is preferred to force, and the smallest instrument that works is chosen.

SHOCKWAVE LITHOTRIPSY

- Introduced in the early 1980s; first successful paediatric SWL in 1986. Guideline role: **URS or SWL** for ureteric stones unlikely to pass; **SWL or URS first-line for renal burden ≤20 mm.**
- Efficacy varies with lithotripter generation, technique, habitus and clearance definition; **habitus matters less in children** — shorter skin-to-stone distance.

PREDICTORS OF SWL SUCCESS AND FAILURE

Factor	Favourable	Unfavourable
Size	<10 mm	>10 mm
Location	Non-lower-pole	Lower pole — adult lower-pole stones <2 cm: PCNL > SWL and URS > SWL , especially 10–20 mm
Density (measure on preoperative CT)	<1000 Hounsfield units (HU)	High attenuation — cystine, brushite
Anatomy	Short infundibulum	Long infundibulum; infundibulopelvic angle >45°

- Nomograms combine prior stone history, location, burden, sex and age.
- **Complications: haematuria expected;** rare subcapsular or perirenal haematoma; fragment colic; **steinstrasse** (ureteric column of fragments), especially after large stones, sometimes needing intervention; **mis-focusing injures colon, vessels, lung, spleen, pancreas;** in adults, SWL for renal (not ureteric) stones is linked to **incident hypertension** — likely a legacy of older wide-focal-zone machines, relevance to children unknown.

CONTRAINDICATIONS TO SWL

Absolute	Relative
Uncontrolled bleeding diathesis	Solitary kidney
Untreated UTI	Urinary tract obstruction
Pregnancy	Abnormal anatomy

Shockwave Lithotripsy Technique in Children

- **General anaesthesia usually** (sedation in some older children); **no bowel preparation**; retrograde opacification occasionally for **radiolucent stones**.
- **Energy matched to size: low power (<20 kV)** → less migration, better fragmentation; **≤2500–3000 shocks per session (<2000 in very young children)**.
- **Shielding and positioning** prevent haematoma and lung contusion; **foam tape shields the lungs** without blocking energy to the stone.
- **Ungated shocks are safe in children** — adult arrhythmias are unlikely.
- **Rate:** in adults, higher rate and energy lower clearance; in children 60 and 90/min were similar, but **80/min cleared better than 120/min** → slower rates favoured.

LARGE RENAL STONES — GUIDELINE VERSUS AUTHORS' PRACTICE

	AUA/EUS 2016	Authors
Renal burden >20 mm	PCNL or SWL (SWL with stent or nephrostomy)	PCNL
SWL upper limit	None stated	No SWL for stones >15 mm
Basis	Expert opinion	SWL poor for large burdens; PCNL safe and highly effective

COMPARATIVE EFFECTIVENESS OF URS AND SWL

Study	Design	Findings	Verdict
Multi-institutional comparison	Retrospective	Similar clearance, residual fragments <4 mm and retreatment; SWL: fewer ED visits (infection, pain) and fewer anaesthetics per treatment	Equal efficacy; SWL less morbid
Randomised trial	Pre-school children	Similar clearance, complications, operating time	Equal
AUA/EUS meta-analysis (unpublished)	Paediatric	Similar clearance	Equal

- **These underpin the guideline: SWL or URS first-line** after failed observation/MET, or for renal burden ≤20 mm.

KEY POINTS: CONSERVATIVE MANAGEMENT AND SHOCKWAVE LITHOTRIPSY

- **Smaller and more distal stones pass more often.**
- **α-Blockers may increase passage** of ureteric stones in children.
- **SWL and URS clear similarly for stones <15 mm**, varying with location, density and anatomy.

Concept — SWL's paediatric advantages (short skin-to-stone distance, fewer anaesthetics, fewer ED returns) are real, but its effectiveness is dictated by stone physics — size, density and lower-pole drainage — so it matches URS only within an envelope of small, soft, favourably placed stones.

PERCUTANEOUS NEPHROLITHOTOMY

- Early reluctance — large instruments, parenchymal damage, radiation, sepsis, bleeding — gave way to experience: **PCNL is first-line for renal stones >20 mm, clearing about 90%.**
- **Bleeding rises with tract size; a 24–30 Fr tract in a child ≈ a 72 Fr tract in an adult.** Miniaturisation → **less bleeding with stone-free rates ≥75%.**

PCNL BY TRACT SIZE

Standard	Mini-PCNL	Micro-PCNL
24–30 Fr	<20–24 Fr; first described 1998, for children; since adopted in adults	<10 Fr; "all-seeing needle" optical puncture; not FDA approved in the US, promising abroad

- Paediatric indications for mini- and micro-PCNL remain under investigation. **AUA allows SWL or PCNL for >20 mm; the authors recommend PCNL, performed only by experienced, trained and equipped urologists.**

Percutaneous Nephrolithotomy Planning for Pediatric Patients

- **Experience is paramount.** CT first → amenability, optimal calyx, size, location, adjacent organs, spinal deformity (scoliosis, lordosis, kyphosis).
- **Consent:** transfusion; **delayed haemorrhage needing angioembolisation;** sepsis; **pneumothorax, haemothorax, urin thorax;** incomplete clearance; adjacent-organ injury.
- **Treat UTI first:** culture with sensitivities **2–3 weeks before;** positive → full course. **Bladder urine may not reflect pelvic urine or stone culture.** In adults, **1 week of empirical ciprofloxacin** may reduce infection. Intravenous ampicillin + gentamicin at operation.

Percutaneous Nephrolithotomy Technique for Pediatric Patients

Sequence: general anaesthesia with **hypothermia prevention** (warm theatre, warmed isotonic irrigant, short operation, draping, temperature monitoring) → cystoscopy + **4–5 Fr open-ended ureteric catheter to just below the pelviureteric junction (PUJ)** → **prone on gel bolsters** (respect contractures; **latex precautions in myelomeningocele;** pad) → **US-guided access** (or fluoroscopy) → **16- or 18-gauge needle** → **No. 11 blade** → **8 and 10 Fr coaxial dilators** → **safety wire, preferably down the ureter** → **18 Fr sheath (24 Fr for staghorn)** → **RPG** → **15/18 Fr rigid nephroscope with dual ultrasonic–mechanical lithotripter or laser** → **flexible scope** to confirm clearance and reach mid-pole calyces → **nephrostomy (7 Fr pigtail in infants → 16 Fr in adolescents)** → removal after urine clears and a **capping trial** → discharge usually **day 1;** **nephrostogram** if the child cannot report pain.

- **Lower-pole access:** the **posterior calyx is the one immediately above the most inferior calyx on anteroposterior fluoroscopy (>90%).** **Upper-pole access:** below the 11th rib, **medial calyx** → less pneumothorax.
- **US target:** the needle appears as a linear echogenic focus; **aim for the papillary tip adjacent to the stone.** Mid-pole mini-PCNL access is feasible for a pelvic stone even in an infant.
- **Paediatric kidneys (especially infants') are mobile and push away** → watch the balloon tip and bony landmarks; **trim the sheath** in small children for manoeuvrability and less torque.
- **Tubeless PCNL: insufficient evidence in children.** "**Sandwich therapy**" (PCNL + SWL for residuals) is historical — limited by PCNL's clearance, miniaturisation and SWL hypertension concerns.

Percutaneous Nephrolithotomy Complications

- **About 10% intraoperative, 20% postoperative;** mostly minor, but **up to 16% ≥Clavien III;** transfusion <10%.
- **Intraoperative:** failure to dilate, **acute haemorrhage, loss of access,** pelvic perforation. **Postoperative:** UTI/sepsis, fragment obstruction, **delayed haemorrhage from pseudoaneurysm or arteriovenous malformation.**
- **Determinants:** operative time, sheath size, mid-calyceal puncture, partial staghorn.
- **Bleeding** → **abort;** **Foley or re-entry catheter through the tract.** **Pelvic injury** → **stop;** **antegrade stent.** Hydrothorax, colonic injury and late bleeding are managed as in adults.

- PCNL only where paediatric interventional radiology can embolise.

KEY POINTS: PERCUTANEOUS NEPHROLITHOTOMY

- Indicated for large upper tract burdens (>20 mm).
- US and/or fluoroscopy for access.
- Clearance about 90%; risks of bleeding, sepsis, pneumothorax, adjacent-organ injury.

Concept — PCNL's place in children rests on one trade: the highest single-session clearance for large burdens, paid for in tract-related bleeding — so every refinement (miniaturisation, US access, trimmed sheaths, interventional radiology back-up) exists to shrink the tract's relative size and its haemorrhagic cost.

LAPAROSCOPIC AND ROBOTIC-ASSISTED PYELOLITHOTOMY

- Not indicated for upper tract stones except with an anatomic abnormality — e.g. pyelolithotomy during pyeloplasty for PUJ obstruction.

SECONDARY PREVENTION

Fluid

- **The cornerstone:** dilutes calcium, oxalate, uric acid and cystine. **Most children drink too little;** adolescents' main barriers are **unawareness of volume drunk and poor response to the need to drink.**
- **Targets:** ≥ maintenance (children); >2–2.5 L/day (adolescents, adults); 1.5–2 L/m² in cystinuria or PH; more with insensible losses.
- **Prevention of Urinary Stones with Hydration (PUSH) trial:** randomised test of financial incentives and coaching to sustain intake.
- **Other fluids:** orange juice, lemonade, blackcurrant juice (↑ pH and citrate) and coffee (↑ volume) protect; grapefruit juice harms; soft drinks controversial. **Counsel water** (sugar, calories) — amount matters more than type.

DIETARY FACTORS

Factor	Mechanism	Recommendation
Sodium	Competes with calcium for tubular reabsorption → calciuria; ↓ salt also ↓ cystine	<2–3 mEq/kg/day (children); <2.4 g/day (adolescents, adults) — hypercalciuria, calcium stones, cystinuria
Calcium	Binds dietary oxalate in the gut → ↓ oxalate absorption	Do not restrict (also osteoporosis); excess vitamin D without calcium supplements not linked to stones
Animal protein	Purines → uricosuria, acid urine; sulfur amino acids → sulfuric acid → ↑ urinary calcium, ↓ citrate and pH; vegetable and dairy protein not lithogenic	100% of daily allowance for age, no excess; in cystinuria little evidence for cystine/methionine restriction, but less animal protein may raise pH
Oxalate	Dietary load; vitamin C → oxalate	Limit in CaOx formers; stop vitamin C supplements if hyperoxaluric
Potassium / citrate	High potassium protective; restriction ↑ calciuria; hypokalaemia ↓ citrate; chronic low intake lowers urinary citrate even without hypokalaemia	Potassium-rich fruit and vegetables
Magnesium	Complexes oxalate in gut and urine; lower risk in males	Supplement may help secondary hyperoxaluria

Factor	Mechanism	Recommendation
Carbohydrate; sucrose	Hypercalciuria; stones	—
Phytate	Binds calcium ; proposed CaOx inhibitor (cereals, legumes, vegetables, nuts)	—

- **The calcium paradox: higher dietary calcium lowers stone risk.**
- **High-oxalate foods:** nuts (almonds, peanuts, cashews, walnuts, pecans), **spinach**, soybeans, tofu, **rhubarb**, beets, sweet potatoes, wheat bran, okra, parsley, chives, black raspberries, star fruit, green tea, **chocolate**.

Medications

- **Drugs when fluid and diet fail. Only thiol agents and lumasiran are FDA approved in children for secondary prevention;** the rest are off-label.

DRUGS FOR SECONDARY PREVENTION

Agent	Indication	Dose: child (adult)	Mechanism	Cautions
Hydrochlorothiazide	Hypercalciuria unresponsive to sodium restriction	1–2 mg/kg/day (25–100 mg/day)	Thiazide → ↓ urinary calcium	K depletion → add potassium citrate
Potassium citrate	CaOx stones, low or normal citrate; uric acid (pH >6.5); cystinuria (pH >7)	2–4 mEq/kg/day (30–90 mEq/day)	Alkali, citrate	GI effects; pH >6.5 → CaP stones
Potassium–magnesium citrate	↓ CaOx recurrence	—	Alkali + magnesium	—
Sodium citrate	Alkali alternative	—	Alkali	Less ideal — sodium ↑ calciuria
Tiopronin (α-mercaptopropionylglycine)	Cystinuria failing fluid, diet, alkali	—	Reduces cystine disulfide → soluble cysteine–drug complex	Commonest thiol; adverse effects similar to or fewer than D-penicillamine
D-penicillamine	As tiopronin	—	As tiopronin	Fever, GI, liver, taste loss, marrow suppression , trace-metal deficiency, membranous glomerulopathy , myasthenia gravis , elastosis perforans serpiginosa

Agent	Indication	Dose: child (adult)	Mechanism	Cautions
Allopurinol	Hyperuricaemia + hyperuricosuria (PRPS, HPRT); adenine phosphoribosyltransferase deficiency; hyperuricosuric CaOx without hypercalciuria, hyperoxaluria or hypocitraturia	4–10 mg/kg/day (max 300 mg/day)	Xanthine dehydrogenase inhibition	Xanthinuria → xanthine stones (rare)
Pyridoxine	PH1	2–5 → 8–10 mg/kg/day	AGT cofactor	Sensory neuropathy at large doses
Lumasiran	PH1	—	RNA interference → ↓ glycolate oxidase	—

- **Thiol-containing agents.** Monitor **liver enzymes, full blood count, urinalysis, copper and zinc; solid-phase assay or high-performance liquid chromatography** distinguishes free cystine from drug complexes, guiding long-term therapy.
- **Allopurinol.** For uric acid stones the mainstay is **high urine flow + alkalinisation**, not allopurinol.
- **Pyridoxine.** **10–30% of PH1 children respond (>30% ↓ urinary oxalate); Gly170Arg or Phe152Ile homozygotes** respond more often and keep renal function, but **sub-genotype does not predict response → trial it in every PH1 patient.** No benefit in other hyperoxalurias unless B6-deficient.
- **Lumasiran.** **ILLUMINATE-A** (randomised trial): **urinary oxalate ↓ 65%; 84% reached ≤1.5 × upper limit of normal.**
- **Urine pH:** target **>6.5 for uric acid, >7 for cystine;** in calcium stone formers on alkali, **avoid >6.5 (CaP).**

Concept — Almost every preventive measure pulls one of three levers — **urine volume, promoter excretion, or inhibitor and pH** — which is why calcium is not restricted (it removes oxalate in the gut), why sodium is (it drives calciuria), and why alkali must stop short of the calcium phosphate threshold.